

Category Theory Reading Club

1st Season: 2026 January-February¹

“Whatever you think maths is ... let go of it now. This is going to be different.”

— (Cheng, 2015)

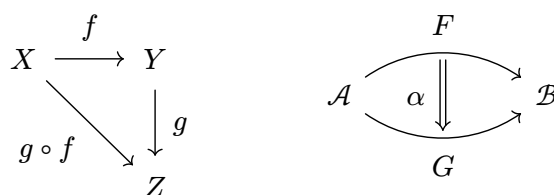
Can you know someone by knowing her friends? Yes, it is common wisdom that persons are more or less defined by their relationships with others. But did you know that there is a famous mathematical result, the [Yoneda Lemma](#), that states and proves exactly this wisdom? The lemma is part of [category theory](#), the *mathematics of mathematics*, the new foundations for mathematics itself, computer science, physics and perhaps intelligence².

If this theory is so cool and important, why you haven’t heard about it? It is considered to be highly inaccessible: you need to learn all of mathematics before learning category theory. This reading club aims to show that this is false. With a *friendly, collaborative, no-questions-are-silly approach*, we will go through introductory material. By sharing our thought processes, we will bring the highly abstract ideas down to earth.

Heavy-weight modern mathematics can be understood in a café style social environment. (Perhaps it is the best way to learn.)

Do you want catch up with the cutting edge of mathematics? but no time for going through hundreds of years of mathematical publications? Then, this reading club might be the right place for you. Come for the shortcut, stay for the discussion.

Category theory has a unique visual way of representing ideas. Only as a sampler, without any explanations, here are two simple diagrams. The left diagram represents *composition*, the right one encodes the more advanced idea of *natural transformations* (again, just take in the visuals, nothing is explained here).



Rules of joining

There is no membership, you can come, listen, and go anytime. No background in advanced mathematics needed.

If you want to join the discussion, we ask you a favour. You read a bit from these books and let us know how it went. Was it possible to understand? What made it difficult? As we try to make the topic accessible, your report will be a highly valued contribution.

¹Check the Infinity Room website for up-to-date scheduling information. <https://egri-nagy.github.io/infinity-room/>

²Speaking of intelligence, we need to dispel a common misunderstanding. AIs cannot answer all mathematical questions. Even if they could, that would not diminish the need for human understanding of mathematical ideas.

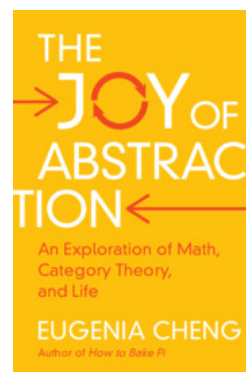
Recommended Books

Barr, M., & Wells, C. (1995). *Category Theory for Computing Science* (Issue v.1). Prentice Hall. <http://www.tac.mta.ca/tac/reprints/articles/22/tr22abs.html>

Cheng, E. (2015). *How to Bake Pi: Easy recipes for understanding complex maths*. Profile.

Cheng, E. (2022). *The Joy of Abstraction: An Exploration of Math, Category Theory, and Life*. Cambridge University Press.

These are available in the library/Infinity Room or online. *How to Bake Pi* (Cheng, 2015) is about recipes, baking cakes, but brings in category theoretical ideas when talking about those. It is a light read in the style of popular science books (or popular mathematics books if that category exists). *Category Theory for Computing Science* (Barr & Wells, 1995) is a classic treatise. Since it has a computing viewpoint, it is a bit more concrete than a pure mathematics book would be. Still, it is a very thick mathematical monograph. Therefore, the best middle ground is possibly the *Joy of Abstraction* (Cheng, 2022), written in a truly friendly style, but containing serious mathematics. For the ease of finding this book, here is the cover.



There was a reading club at the Topos Institute, where the author herself posted explainer videos. <https://topos.institute/events/joa-bookclub/> For a collection of definitions, see her lecture notes: https://egri-nagy.github.io/infinity-room/DOCS/2002_categorynotes_cheng.pdf or <https://math.ucr.edu/home/baez/qg-winter2016/CategoryTheoryNotes.pdf>. These notes, of course, are unreadable. They should be used for referencing purposes, almost like a dictionary.

For more information contact [egri-nagy](mailto:egri-nagy@aiu.ac.jp) at [aiu.ac.jp](mailto:egri-nagy@aiu.ac.jp).

FAQ

But what is category theory? Is there a short definition?

Category theory is very abstract and (consequently) it applies to many things, therefore it is difficult to give a short but meaningful definition. This quote comes close to that.

“In the end, mathematics is all about finding similarities between things, and category theory is about finding similarities between mathematical things.”

— (Cheng, 2015)

For instance, numbers capture the idea that collections can be similar in having the same number of things. Two apples and two bananas – forgetting the details leads to the abstract concept of two.

Both studying shapes (topological spaces) and programming languages both can be done with the tools of category theory. Therefore, category theory is *a general theory of mathematical structures*.

Why learning about category theory?

“What’s the point?” – this is a very good question. We should be asking this more often – when doomscrolling or binge-watching low quality videos... But back to mathematics, the effects are

not direct. You will never use the Adjoint Functor Theorem in everyday life. However, you use your thinking skills all the time, and working with abstract ideas sharpens your intuition.

“I am convinced that my years of stretching my brain in those abstract ways enabled me to think more clearly about the world around me, and more easily make connections between situations, connections that others don’t see.”

— (Cheng, 2022)